IONIC EQUILIBRIUM

1. log 2 = 0.301
log 3 = 0.477
log 7 = 0.85

2. pH of → a) Weak Acid

↳ simple diss $HA \rightleftharpoons H^+ + A^-$

$C - C\alpha \quad C\alpha \quad C\alpha$

$$\alpha = \sqrt{\frac{K_a}{C}} \Rightarrow [H^+] = \sqrt{K_a C}$$

$$\boxed{pH = \frac{1}{2}(pK_a - \log C)}$$

b) WA + SB

↳ Strong B. eg, NaOH, WA eg. CH_3COOH ko pura khaa jaega aur beaker mein rahega

Depending on the amount of WA & SB ←
- → $CH_3COOH + CH_3COONa$
- Ya → $CH_3COONa + NaOH$ ⇒ calculate the pH using NaOH only
- Ya → only CH_3COONa ⇒ Hydrolysis maardo

$[CH_3COO^- + H_2O \rightleftharpoons CH_3COOH + HO^{\ominus}]$

$K_a \times K_b = K_w$

↳ $K = K_b = \frac{K_w}{K_a}$

And, $CH_3COOH + CH_3COONa$ → {diss maaro in presence of C.B}

↳ $CH_3COOH \rightleftharpoons CH_3COO^{\ominus} + H^+$

$C \quad C' \quad -$

$C - C\alpha \quad C' + C\alpha \quad C\alpha$

$$\boxed{pH = pK_a + \log\frac{[Salt]}{[Acid]}}$$

Henderson Hasselbalch eqn

c) WA + WB

↳ Salt banao eg. $\underset{c}{CH_3COONH_4}$

Ab eq. pe yehi conc fixed hain →

$$\underset{c-ch}{CH_3COO^{\ominus}} + \underset{c-ch}{\overset{\oplus}{NH_4}} + H_2O \rightleftharpoons \underset{ch}{CH_3COOH} + \underset{ch}{NH_4OH}$$

: ab ~~try~~ diss ya hydrolyse karado

kisi bhi eqⁿ se pH nikal skta hai!

$$\underset{c-ch}{CH_3COO^{\ominus}} + H_2O \rightleftharpoons \underset{ch}{CH_3COOH} + HO^- \checkmark$$

nikaal lo

$$\underset{c-ch}{\overset{\oplus}{NH_4}} + H_2O \rightleftharpoons \underset{ch}{NH_4OH} + H^+ \checkmark$$

$$\underset{ch}{CH_3COOH} \rightleftharpoons \underset{c-ch}{CH_3COO^{\ominus}} + H^+ \checkmark$$

$$\underset{ch}{NH_4OH} \rightleftharpoons \underset{c-ch}{\overset{\oplus}{NH_4}} + HO^{\ominus} \checkmark$$

$$\boxed{pH = 7 + \frac{1}{2}(pK_a - pK_b)}$$

3. Dibasic / Tribasic acid mein

↳ $K_{a_1} \gg K_{a_2} \gg K_{a_3}$

↓

use to calculate pH.

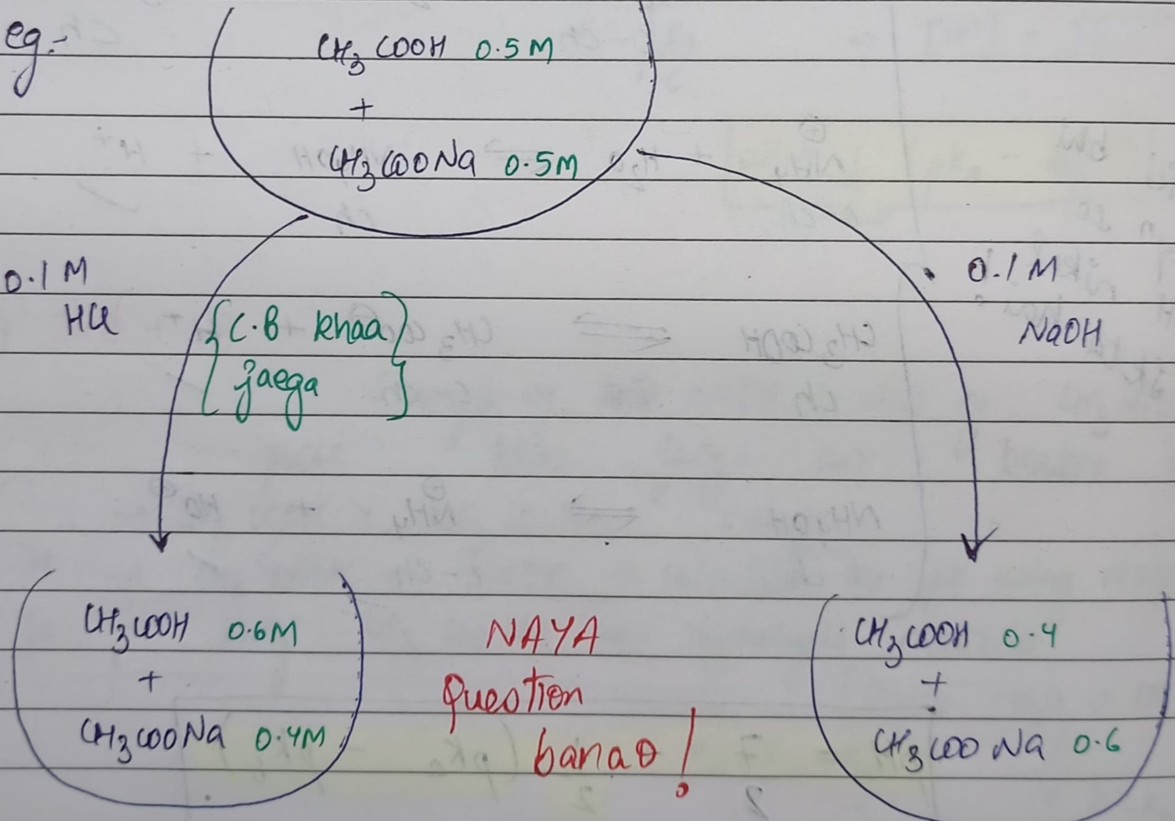
4. Strong Acid ⇒ 10 times dil^n ⇒ pH → ① ↑

Weak Acid ⇒ 10 times dil^n ⇒ pH → ⓪.⑤ ↑

5. Buffer → Dono acid / base ko khaa skta hai.

$$\boxed{pH = pKa + \log \frac{[Salt]}{[Acid]}}$$

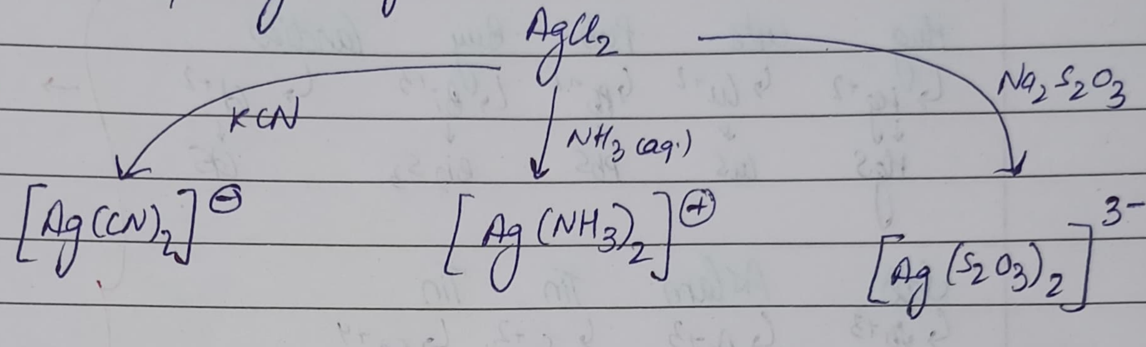
eg:-



6. Solubility

$AgCl$ → white ppt, soluble in aq. NH_3
$AgBr$ → pale yellow ppt, part. soluble in aq. NH_3
AgI → Yellow ppt, × × ×
$PbCl_2$ → white ppt (soluble in hot H_2O)
$Al(OH)_3$ → white glatinous ppt
$Ca_3(PO_4)_2$ → white ppt

Factors : ① Common ion effect
② Complexing Agent



③ pH of solⁿ
↳ Agar salt ka koi ion hydrolysis mein jaata hai

* ppt kab banega?

Q ⇒ Ionic Pro.

① IP $< K_{sp}$ → no ppt
② IP $\geq K_{sp}$ → ppt starts

④ Temp.
↳ K_{sp} hi change ho jaega

Generally salts ka diss → endo
$\therefore T\uparrow \Rightarrow K_{sp}\uparrow \Rightarrow s\uparrow$
$\longleftrightarrow$

{* $Ce_2(SO_4)_3$ ↳ exo $\Rightarrow T\uparrow \rightarrow S\downarrow$ }

7. Sulphides

Hug ↳ Hg^{+2} ↓ HgS
Cute ↳ Cu^{+2} ↓ CuS
Pub ↳ Pb^{+2} ↓ PbS
Buy ↳ Bi^{+3} ↓ Bi_2S_3
Candies ↳ Cd^{+2} ↓ CdS

Sab ↳ Sb^{+3} ↓ Sb_2S_3
Aslam ↳ As^{+3} ↓ As_2S_3
Tin ↳ Sn^{+2} ↳ SnS
Tin ↳ Sn^{+4} ↳ SnS_2

} → Acidic medium mein ppt banate hain
⇓
K_{sp} ↓↓

Ni Ni^{+2} ↓ NiS
Ko Co^{+2} ↳ CoS
mann Mn^{+2} ↳ MnS
Zann Zn^{+2} ↳ ZnS

} → Basic medium
↳ K_{sp} ↑↑

8. Indicators

Methyl orange – 3.1 – 4.5
Phenolphthalein – 8.3 – 10
Bromothyl Blue – 6 – 7.5

Indicator → { [HIn] } W.A

$$HIn \rightleftharpoons H^+ + In^-$$

H^+ Beaker mein pehle se hai ⇒ Iss $[H^+]$ ke hisaab se $[HIn]$ aur $[In^-]$ ka ratio aega.

✓ colour
When $[HIn] = 10[In^-]$

When $10[HIn] = [In^-]$ ✓ colour

$$\boxed{pH = pK_{IN} + \log\frac{[In^-]}{[HIn]}}$$

Ye beaker batata hai

Ye pH ke hisaab se aega